Home Assignment – Clean Air Finder

The business case

A relocation service helps families decide where to live in Bulgaria. Their most common question is: **"Which Bulgarian city had the cleanest air over a given period?"**

They want a simple tool that answers it from real measurements, not mock data.

What to build

A web service and a minimal web UI.

1. The user picks a **historical time period** (from – to, using a calendar).
2. On a button press, the app shows the **city with the cleanest air** over that period, with its average air quality and enough context for a non-expert to trust the result.
3. The app works with **real, publicly available data**. Finding the sources is up to you. They must be reachable through a public API, open downloads or free registration, and their terms must allow your use.
4. Data is **collected and stored locally** by your app and **kept up to date**. Do not proxy each user request to a third party.
5. The stored history must cover at least the **last 12 months**, and any period within it can be queried.
6. Any data older than the last 36 months should be archived to save storage costs.
7. Invalid or impossible requests get a clear, useful response.

How it should behave

- **Fast.** Queries stay fast whatever the city or period length. Show us your process for measuring and deciding performance with the tools you prefer.
- **Production-ready.** Treat it as something you'd hand to an on-call team.
- **Resilient.** A third-party source being slow, down or returning bad data must not break the app.
- **Deployed or deployable.** Use any platform, with a single documented command to run it.

- **Observable.** Use an observability platform of your choice. We should be able to tell whether the app and its data collection are healthy.
- **Storage is your call.** A database engine is optional. Choose what fits and justify it.

What to send us

A link to a Git repository containing:

- The source code, with a commit history that shows how the work progressed
- A **README** covering how to run it, the architecture, your data sources and why you chose them, your assumptions, the trade-offs you made, and what you'd do with more time
- A live URL, if you deployed it
- An **AI usage note** (see below)

Using AI

AI tools are welcome. Use them the way you would at work.

In a short note, tell us what you used them for, where they helped, and where you disagreed with or corrected them. You should also tell us how you used AI for this task, and what your workflow looks like. You own every line you submit, so be ready to explain and change any of it.

What happens next

Your submission is the starting point for a technical interview. You'll walk us through your decisions, and we'll extend the app together.

We care most about clear thinking and sound judgment, then correctness, performance and operability. Polish matters less. A focused, well-reasoned solution beats a sprawling one.